

Automatic Water Dispenser

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Table of Contents

1	Introduction	6	Implementation Steps
2	Objectives	7	Arduino Code(Snippet)
3	Components Used	8	Applications
4	Circuit Connection	9	Conclusion
5	Working Principles		



Introduction

Automatic water dispensers provide a contactless and hygienic method for dispensing water. This project uses Arduino UNO, HC-SR04 ultrasonic sensor, and a servo motor to simulate a smart tap system. Water is dispensed automatically when a hand is detected.

Objectives

- Develop a contactless water dispensing system
- Use ultrasonic sensing for hand detection
- Control dispensing using a servo motor
- Ensure efficient, hygienic water delivery

Components Used

1. Arduino UNO

A 5V microcontroller board (ATmega328P) with 14 digital I/O pins, used to control components like sensors and motors in embedded systems.

2. HC-SR04 Ultrasonic Sensor

A distance sensor with a 2–400 cm range (± 3 mm accuracy), emits ultrasonic waves and detects echo to measure proximity.

3. SG90 Servo Motor

A 4.8V–6V micro servo with 0°–180° rotation, controlled by PWM signals for precise angle movement in lightweight automation.

4. Jumper Wires

Flexible, solderless wires (male/female) used to connect electronic components on breadboards or between modules..

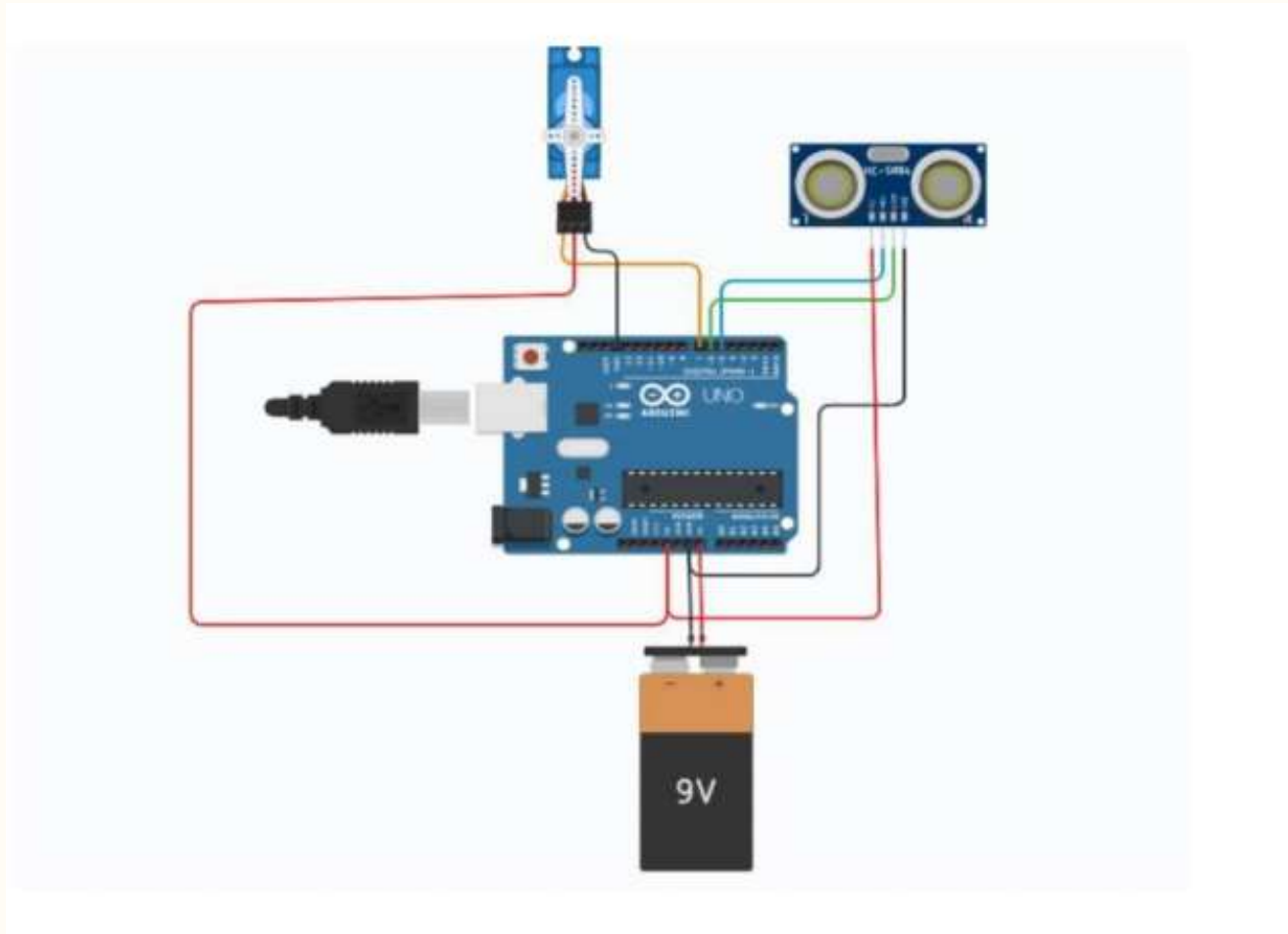
5. 9V Battery with Clip

A portable 9V power source for Arduino boards, regulated to prevent overvoltage in standalone applications.

6. Breadboard

A solderless prototyping platform to organize and connect electronic components safely and efficiently.

Circuit Connection



Working Principle

Connect the components as per the diagram

Upload the Arduino code via USB

Open Serial Monitor (9600 baud)

Bring glass near the sensor

Servo opens and closes simulating water flow

Implementation

- Connect the components as per the diagram
- Upload the Arduino code via USB
- Open Serial Monitor (9600 baud)
- Bring glass near the sensor
- Servo opens and closes simulating water flow

Arduino Code (Snippet)

```
#include <Servo.h>
#define trigPin 10
#define echoPin 11
#define servoPin 9

Servo waterServo;
void setup() {
  Serial.begin(9600);
  waterServo.attach(servoPin);
  pinMode(trigPin, OUTPUT);
  pinMode(echoPin, INPUT);
  waterServo.write(0);
}
```

```
void loop() {
  float distance =
  measureDistance();
  if (distance > 0 && distance
  < 10) {
    waterServo.write(90);
    delay(1000);
    waterServo.write(0);
    delay(1000);
  }
}
```

Applications

- Public washrooms
- Hospitals and clinics
- Restaurants and kitchens
- Smart homes

Conclusion

This project successfully demonstrates a simple and effective automatic water dispenser system. It highlights the use of sensors and actuators in real-world automation solutions, encouraging further development in contactless technology